Installation and Use Manual

FC-280 Series

Mass Flow Controllers and Flowmeters

Mass Flow Controllers
FC-280A, FC-280AD, FC-280SA , FC-282S

Mass Flowmeters
FM-380A, FM-380AD

Mykrolis

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SYSTEM DESCRIPTION

Tylan General Mode FC-280 Series Mass Flow Controllers Accurately and reliably measure and control the mass flow rate of gases. They have been specifically designed to Allow operation on any gas having a known molar specific heat.

The FC-280 series of mass flow controllers and flowmeters Include:

Mass Flow Controllers

Low Flow (10 sccm to 30 slm full-scale)

FC-280A	Fast response of less than 2 seconds
FC-280AD	Integral 3-1/2 digit display
FC-280AJ	4-20 mA Input and output
FC-280SA	Ultra-fast response of 400-800 ms
FC-280SA-P	Ultra-fast response with valve drive signal on Pin D

Medium Flow (50 slm to 200 slm full-scale)

FC-282S	5 seconds response

Mass Flowmeter

FM-380A	10 sccm to 30 slm
FM-380AD	Integral 3-1/2 digit display

The low flow instruments have a user-adjustable range from 10 sccm to 30 slm full-scale flow rate. This is accomplished in the flowmeter section by adjusting the bypass. In the controller section, many changes in flow

Range can be accomplished by an external adjustment of the control valve alone. Very large range changes may involve replacing of the valve orifice.

The instruments operate on ±15.0 VDC power, provide a 0 to 5.0 VDC indicated output signal linearly proportional to the flow rate, are RFI/EMI protected and are both mechanically and electrically interchangeable with the tylan general FC-260 series flow controllers. The “J” option provides 4-20 mA control and output and is available when specified at tine of order. An adaptor fitting is available for interchangeability with the tylan general FC-261 series instruments. Other inherent design features include:

- Enhanced stability of both zero and span
- Insensitive o mounting position
- Fast response with minimal overshoot
- External electrical valve override
- Normally closed solenoid valve with both mechanical and electrical “purge” capability
- Ease of maintenance

PRINCIPLE OF OPERATION

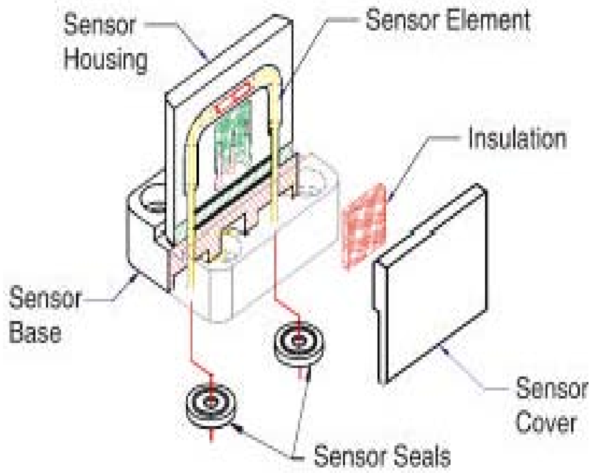
The flow controller is a self-contained, closed-loop control system which measures the mass rate of gaseous flow through the instrument, compares this with an externally commanded flow rate, and adjusts the valve to control the flow to the commanded level. The flow controller consists of four basic elements which accomplish this function:

- The flow sensor
- The bypass, or flow-splitter, which determines the full-scale flow range of the measuring section
- The control valve
- The electronics which condition the flow signal and drive the control valve

Flow Sensor

The flow sensor consists of two self-heated resistance thermometers wound around the outside diameter of a thin-walled capillary tube. These coils, each having a resistance of 330 ±13 ohms at 20°C, are connected in a bridge circuit and supplied with a regulated current. The heat generated by the power dissipated in the coils raises the tube temperature approximately 70°C above ambient. At no flow, this heat is symmetrically distributed along the tube. With gas flowing in the sensor tube, heat is carried downstream. The resulting shift in temperature makes the upstream sensor cooler than the downstream sensor. This temperature difference (and corresponding electrical resistance difference) is directly proportional to the mass flow rate of the gas through the tube.

The bridge output, being a direct function of the resistance difference, is amplified and further linearized by the electronics to give a 0 to 5.0 VDC indication of flow rate. Increasing the flow rate well above the full-scale range of the instrument will eventually cool the entire sensor tube and the output signal will reverse and asymptotically approach zero.



The capillary tube is dimensioned to have a minimal mass (for fast response) and an extremely large length-to-diameter ratio to ensure laminar flow over the full operation range. The optional large bore sensor tube is designed for low vapor press gas applications where minimal pressure drop is required, but is limited to full scale flow rates of 3 slm or less.

The housing which encases the sensor tube is precisely configured to minimize natural convection natural convection currents from one coil to the other, thus allowing the instrument to be mounted in any position with no zero adjustment required to re-establish the original calibration. This configuration minimizes the mass of the sensor resulting in a time constant that is one-third that of other sensors of similar design. Over flow controller response can therefore be dynamically controlled to eliminate overshoot.

Bypass(Flow-splitter)

The FC-280A/FC-280SA/FM-380A bypass, which is located in the primary flow path in the base assembly, produces a linear pressure drop versus flow rate between the inlet and outlet of the sensor tube, which in turn produces a 0 to 100% sensor flow for 0 to 100% flow of the instrument. In order to ensure a constant ratio between sensor flow and total flow (independent of pressure, temperature, and gas properties), the bypass arrangement has been designed to maintain the flow well within the laminar region of fluid flow over the entire range of the instrument.

The bypass consists of a tapered flow restrictor concentrically located in the tapered bore of the base. The resulting annular flow path develops the required pressure drop versus flow rate relationship necessary to drive the sensor flow through its full operating range. Axial adjustment of the restrictor in relation to the bore varies the thickness of the annular gap and thus provides a means for continuous adjustment of the full-scale range of the instrument from 10 sccm to 30 slm.

The advantages of this patented configuration are as follows:

- Excellent agreement between actual and theoretical gas conversion factors is achieved, allowing the instrument to be used with any gas by simply varying the electrical gain in direct proportion to the theoretical conversion factor.

- A wide adjustment range is provided within the same instrument eliminating the need for pars replacement in order to change the flow range.

- Accuracy degradation due to contamination build-up is greatly reduced due to the large surface area of the restrictor and bore.

- The restrictor can be easily removed and cleaned during routine maintenance procedures.

The FC-282S bypass consists of a variable area, cylindrical screen bypass which produces the necessary linear pressure drop to drive the sensor through its operating range. This configuration is necessarily different from the annular FC-280A/FC-280SA bypass to accomm- odate the higher flow rates of the FC-282S.

Valve

The control valve is a normally-closed, solenoid valve which is extremely fast, easily adjusted, and has 2% or better shutoff capability. Except for the dynamic plunger, all gas wetted metal surfaces are of 316 stainless steel. The plunger is 446 stainless steel.

The range of the valve is determined by the size of the control orifice, which is screwed into the controller base. These precision valve seats are 316 stainless steel and provide excellent control resolution by engagement with the teflon face of the valve poppet. The valve seats are easily replaceable for cleaning and instrument range changing.

The valve poppet is uniquely designed to provide a relatively constant control range (stroke) for each size of the valve seat. different ranges within the full range of a particular valve seat are accommodated by the amount of voltage applied to the valve.

This arrangement provides consistent dynamic response and static stability with only a minimal need for electronic “tuning.”

An adjustment on top of the valve is provided to allow for setting the relative position between the valve poppet and valve seat following re-assembly after cleaning or seat replacement (see Adjustment Procedures) this adjusting nut may also be used to mechanically open the valve for purging purposes in the event of a power failure or an instrument malfunction, which may necessitate removing the instrument from the gas system.

Electronics

To ensure intrinsically safe operation, the circuit has been designed in accordance with the basic requirements of ISA-RP12.2, in the that there is no inductance (except in the solenoid valve itself) and the capacitance values are below the level which will provide enough energy to create a spark under foreseeable fault conditions. The higher capacitance elements are in series with large resistance values, which provide the necessary current limiting protection.

The circuit consists of three interrelated sections: The bridge supply circuit consists of an operational amplifier which supplies the flow sensor bridge with a constant current of 13.0 milliamps as determined by the reference voltage. The bridge current may be increased to the 15.0 milliamp level required for operation with the large bore sensor. Linearity results are fed back from the flowmeter output to alter the bridge current as a function of the output with linearity adjustment potentiometer adjusting the magnitude and sign (positive or negative) of this feedback. This linearity control has its maximum effect at 50% output and little or no Effect at 0% and 100% output, resulting in virtually independent zero, gain and linearity adjustments.

The flowmeter amplifier circuit amplifies the bridge output to give a 0 to 5.0 VDC output signal linearly proportional to flow rate. The gain adjustment pot provides a 20% gain adjustment, and the zero adjustment pot provides a ±30% zero adjustment.

The speed-up circuit (FC-280SA/FC-282S only) provides the necessary frequency response correction for the output signal. This results in expanding the sensor cutoff frequency by about one decade, which is equivalent to "speeding-up" the sensor response time. P4 allows for tuning to the original sensor cutoff frequency.

The setpoint signal conditioning circuit limits the speed of setpoint changes. A step change of setpoint is converted to a signal with tapered edges enabling the controller to follow the setpoint without large control deviations at any time.

The controller circuit generates the valve drive signal as a function of Output and setpoint signal. First, the control deviation is computed. This signal is then processed by the true P-I-D controller in order to drive the control valve using a transistor. Control parameters are influenced by proportional band, integral action, and differential action. The valve protection circuit disables the valve driver at setpoints below a threshold voltage of 50 mV. Additionally, there is an external setpoint override capability through pins D and L Pin D is internally pulled low to -14V. Grounding this pin to common overrides the setpoint and closes the valve. Similarly, TTL input 'low ' at pin L effectively disables the setpoint and closes the valve.

NOTE: With "P" option (FC-280SA/FC-282S only), grounding pin D opens the valve to maximum, see Figure 13.

Cover

The cover on the instrument is designed to accept both round and flat cable mating connectors in a single configuration. The cover is made of high strength plastic and is copper-nickel-chrome plated to provide 60db (minimum) attenuation of RF noise. This, in conjunction with the eight roll-off and filter capacitors in the electronics, provides excellent EMI protection.

INSTALLATION

For maximum performance and service life, the instrument should be installed in a clean, dry atmosphere, relatively free of shock and vibration. Sufficient room for access to the electronics and plumbing should be provided to facilitate maintenance and removal for cleaning. Fitting dust caps should not be removed from the instrument until installation.

Gas Connections

The various types of fittings available are shown in Figures 3 and 4. Tubing should be pre-cleaned and polished to eliminate particulate contamination and ensure leak-tight operation. After installation of the instrument and prior to its use, the plumbing system should be thoroughly leak-tested and purged. Since the control valve in the instrument is not guaranteed to provide positive shut-off, it is recommended that a separate shut-off valve be installed in series (upstream to eliminate a flow surge during turn-on downstream to prevent back migration of contaminants in to the control valve or both).

When the FC-280A/FC-280SA is used to replace the FC-261, an outlet adaptor kit (Tylan General P/N 905516001) is available to extend the overall length of the flow controller, matching the length of the FC-261.

Electrical Connections

Refer to the appropriate Electrical Hookup Diagram.

POWER : Any ±15 VDC power supply meeting the requirements as designated in the specifications may be used to energize the instrument.

CONTROL SIGNAL : Any 0 to 5.0 VDC command voltage having a source impedance of 2500 ohms or less may be used. The input impedance of the flow controller is 1 megohm (minimum). 4-20 ㎃ current option is also available. Contact factory for details.

OUTPUT INDICATION : Any 0 to 5.0 VDC meter with at least 1000 ohms/volt can be used to provide visual indication of the mass flow rate. Recorders, voltage dividers (for conversion to engineering units) and other instrumentation may be added, provided the total load impedance is mo less than 2000 ohms. The source impedance of the flowmeter output signal is less than 1 ohm. The FC-280AD/FM-380AD contains a built-in display which may be used to read the output and control signals.

NOTE : Exercise caution when connecting the instrument to currently available readout boxes and calibrators which are manufactured by Unit Instruments, Inc. This equipment applies 15VDC to pin E (FC-280SA/FC-282S valve test point). With jumper option “P” (see Figure 13) the valve test point is moved to pin D, which makes it compatible with Unit Instruments’ test sets but no longer retrofittable with the Tylan General FC-260 series soft-start function.

Integral Display (FC-280AD/FM-380AD only)

The FC-280AD/FM-380AD features a 3-1/2 digit liquid crystal display mounted in the side of an otherwise standard FC-280A/FM-380A. Additional power consumption for the 1.25" X 0.5" display with 1/2" high is negligible.

During normal operation the display shows the indicated flow rate. Depressing a momentary push button on the side of the instrument switches the display to read the commanded setpoint. The display may be set to read in engineering units (sccm or slm) or as a percentage of full scale.

Environmental Requirements

Refer to Figure 9- Specifications.

TEMPERATURE : The instrument may be operated at any temperature between 0°C and 50°C, provided the gas and ambient temperatures are maintained equally. Since the indicated flow rate has a temperature coefficient of ±1 % per °C, the user may want to calibrate the instrument at the actual operating temperature to maximize the measurement accuracy.

PRESSURE : Flow controllers may be operated at any gas pressure up to 150 psig (1135 kPa). Since the indicated flow late varies in direct proportion to specific heat (Cp), which varies differently with pressure and temperature depending upon the molecular structure of the gas, it is recommended that the instrument be calibrated at the actual operating pressure.

NOTE : The pressure coefficient of ±0.007 % per psi (±0.001 % per kPa) generally applies to monatomic and diatomic gases only.

The flow controllers, which can be pressurized to 1500 psig (10,500 kPa) without damage, have a maximum operating pressure of 150 psig and the differential pressure (inlet to outlet) must be maintained within the specified 10 to 40 psid (70 to 275 kPa) range. Lower differential pressure ranges are available on special order.

The flow controllers may be operated into a vacuum as long as the inlet pressure is 15 psia or greater. Lower differential pressure ranges are available on special order.

Calibration

Each instrument is factory calibrated for the specific flow range and gas indicated on the nameplate. Standard factory calibration is within ±1.0% and is referenced to standard temperature and pressure * The calibration for other gases can be approximated to ±5% using the conversion factor charts. Factory calibration utilizes the gases on Gas Flow Conversion Factor Charts. Calibration checks with other gases can show discrepancies of up to ±5%.

To obtain calibration accuracy of ±1% after range change or for other gases, precision calibration equipment will be required. As detailed in the Adjustment and calibration procedures section, range changes can be made to within ±10% by removing the inlet fitting and adjusting the bypass. Fine tuning to the desired accuracy level can then be accomplished by adjusting the potentiometers on the printed circuit board in conjunction with a reference flow standard.

NOTE : In accordance with SEMI Standard E12-91, Standard Pressure and Temperature are defined as 760mmHg and 0°C respectively.

CAUTION : An instrument originally calibrated for a non -reactive gas should not be converted for use with a reactive gas unless all of the seals are replaced with a suitable compound. Additionally, any instrument which has been in reactive or corrosive gas service should be thoroughly purged and cleaned prior to conversion to another gas. Instruments are factory assembled using Viton for non reactives, Neoprene for Ammonia, and Kalrez for all other reactive and corrosive gases unless specified otherwise. Refer to spare parts O-ring Kits.

Mounting

Two 8-32 UNC tapped holes are provided for mounting. The instrument is essentially insensitive to mounting position. Refer to Figure 16 for mounting configuration.

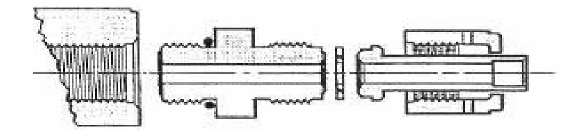


Figure 3
Vacuum Gas Fitting (With Flat Gasket)

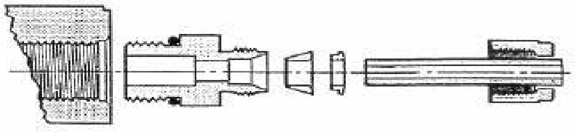


Figure 4
Compression Gas Fitting

OPERATING INSTRUCTIONS

Installation and Start-up

The interconnect cable should be tested separately for continuity, pin-to-pin shorts, and correct pin assignments per the electrical hook-up diagram shown in Figure 10. The insulation resistance between pin 2 and the base of the instrument should be verified to exceed 50 megohms at 50 VDC, and the ground connection between pin 1 and the base should be verified to be less than 1 ohm. After engaging the mating connector, secure it to the cover with two 4-40 X 3/4 inch long screws. Do not over-torque. After installation in the system and prior to operation, any reactive gas system should be thoroughly leak checked and then purged with dry nitrogen to eliminate the presence of air and moisture. Because the instrument is equipped with a normally-closed control valve, it cannot be purged unless the valve is commanded to open. This can be done electrically in two ways. With ± 15 volt power applied, apply either a full-scale(5 VDC) setpoint or ground the valve test point (normally pin E, with "P" option, pin D).

A "cycle" purging technique is more effective in removing atmospheric contaminants than a simple continuous purge gas flow. To cycle purge, alternate the flow of purge gas with a pump down of the gas system to vacuum for several cycles. If vacuum is not available, reducing the pressure to zero psig for several cycles is recommended. Cycle purging helps remove contaminants from small, blind cavities in the system which together constitute a virtual leak source. A more detailed procedure can be found in Appendix B.

Apply ±15 VDC power and allow a 30 minute warm-up time before pressurizing the system with process gas. If the indicated output does not settle to zero to within ± 1% full scale (or the level of zero desired), re-zero the instrument (see Adjustment Procedures) before proceeding. Once the zero has been verified, pressure may be applied. After establishing that no hazardous condition will be created by the venting of process gas, flow controller operation may then be verified over the complete operating range. If there is any question that gas system pressures may not be correct, verify operation with the following procedure: If the flow output voltage voltage is within a few millivolts of the setpoint, the flow controller is controlling properly. Check for this condition at the highest and lowest flows anticipated at both the highest and lowest input-output pressure differentials anticipated.

Safety Features and precautions

Every effort has been made in the design of the instrument to provide safe, trouble-free operation. Key features include reverse power supply polarity protection, output over-voltage and short circuit protection, low component temperatures, and conformance to intrinsic-safety design criteria. Additionally, a built-in "threshold" circuit ensures that the valve is commanded closed at zero command independent of any existing output offset, and the valve automatically closes in case of a power failure.

Two other features are provided for the operator's convenience during start-up, abnormal, or fault conditions. First, the valve command circuit can be overridden externally to either drive the valve fully open or fully closed. Connecting pin E to 0 VDC will drive the valve open, while connecting pin D to 0 VDC will drive the valve closed. A second input to close the valve is provided on pin L. This input is active low and conforms to TTL Logic levels. Second, should there be a need to purge the system during a power failure or instrument malfunction prior to removing the instrument from the system for repair, the valve may be mechanically opened by first removing the instrument cover and then turning the adjusting nut on top of the valve one or two turns counterclockwise. Readjustment of the valve after purging or repair is described in the Adjustment Procedures.

The following precautions should be taken by the user to prevent damage, minimize safety hazards, and maximize performance.

1. Use pre-cleaned tubing, free of particulate contamination.
2. Thoroughly leak test the entire system prior to operation (recommended level is 1X10-9 cc/sec of helium or less)
3. Use only clean, moisture-free(dry) gases.
4. Purge only with dry nitrogen (or other inert gas) before and after breaking into the system. Connecting pin E to 0 VDC will readily allow for this by driving the valve open.
5. DO NOT purge reactive gas systems with inert gases between runs unless it is required by the process. Even “dry” gases contain some amount of moisture which will result in contamination build-up.
6. NAVER insert or unplug the connecting cable or the control valve leads with power on if there is a possibility that the ambient atmosphere might be explosive.
7. ALWAYS command no flow or ground pin D or pin L when the gas supply is shut off. The command signal should be interlocked with a series shutoff valve (as shown in Figure 10) to prevent unnecessary over-heating of the control valve during shutoff, and excessive flow surges after turn-on.
8. Avoid installation of the instrument in close proximity to high sources of RF noise and / or mechanical vibration. If this is unavoidable, proven methods of instrumentation filtering cable shielding, and/ or shock mounting should be utilized.
9. See Caution Note on page 6 before converting any instrument for operation on a gas other than what is was originally calibrated for or previously used on.

Operating Modes

In general, the instruments may be operated at any pressure and temperature that falls within the limits stated in the specification. Optimum performance, however, can be achieved and will prevail if the operating pressure and temperature are pre-determined and controlled to a narrow range. Calibrating and “fine tuning” the instruments at the actual operating pressure, temperature and flow rates can significantly improve the performance characteristics.

The instruments are factory calibrated at an ambient temperature of 23°C ±13 0°C and 30 psig ± 10 psig (315 kPa ±70 kPa) inlet pressure and corrected to standard conditions. Flow controllers are adjusted to pass full rated flow at 8 psig (160 kPa) inlet pressure (with 0 psig outlet pressure) and shut down to less that 2 % of full scale at 40 psig (380 kPa) inlet pressure. Response time is verified at both extremes.

In vacuum systems, flow controllers maintain their calibration accuracy due to the pressure drop of the control valve downstream of the flow sensing section. The increased pressure drop of the valve due to the increased gas velocity reduces the full-scale flow rate and unless there is sufficient inlet pressure available to overcome the increased pressure drop required, increasing the range of the instrument and/or re-adjustment of the valve and /or the valve voltage may be necessary.

Since the control valve is normally closed, it is not necessary to provide an external time-delayed "soft-start" signal provided the command signal is maintained at zero during the "off" mode and is then applied to the instrument at the same time (or after) pressure is applied. Alternately, pin D connected momentarily to ground prior to applying gas pressure also prevents excessive flow transients at turn-on. A slower ramp change may also be selected for vacuum systems which have small volumes and large flow rates or in other applications where a slower ramped change in flow rate is desirable.

Flow controllers may be operated at inlet pressures of up to 150 psig (1150 kPa), provided the differential pressure is maintained within the specified range for the full scale flow rate (see Figure 9 Specifications) and the outlet pressure of the flow controller is regulated to a relatively constant level. Flow controllers can withstand pressure surges of up to 1500 psig (10,500 kPa) without damage, but are not designed to operate in a controlling mode above 150 psig (1150 kPa) inlet pressure.

MAINTENANCE, ADJUSTMENT AND RANGE CHANGE

Routine and Preventative Maintenance

It is recommended that a routine maintenance schedule be established on each instrument in order to maximize its useful service life. This would include, as a minimum, cleaning and recalibration. The optimum (or necessary) service period being dependent on usage, environ-mental conditions, gas corrosiveness, etc., must be established by the user based upon historical experience in that particular application.

The recommended steps for routine and preventative maintenance are as follows :

1. Purge the instrument with dry nitrogen for a minimum of 30 minutes prior to removal from the system.
2. Verify the calibration of the flow metering section by comparison with a suitable reference standard or calibrator.
3. Operate the instrument in the control mode over its entire operating range (2 to 100 % full scale) at the minimum and maximum inlet pressures of 10 and 40 psig(380 kPa). Check response, stability and control resolution.
4. Remove the inlet and outlet fittings and visually examine for signs of contamination. Examine the seals for any evidence of hardening, swelling or contaminant accumulation.
5. Based upon the results of steps 1 through 4 above, either reinstall in the system (followed by a nitrogen purge) or proceed to the

applicable Cleaning, Adjustment and/ or Calibration Procedure described in this manual.

Cleaning

Should the instrument show symptoms of internal flow path contamination, it should be disassembled and cleaned. The seals on the fittings, sensor and valve should be examined for hardening, swelling, inflexibility, and / or contamination accumulation and should be replaced if any of these symptoms prevail relative to new and unused seals. The recommended sequence of solvents used for cleaning (either for flushing the assembled instrument or for soaking individual components in an ultrasonic cleaner) is as follows:

For instruments used with all gases except those containing silicon or germanium compounds:

a. Distilled water (2 to 3 minutes)
b. Alcohol (2 to 3 minutes)
c. Blow dry with dry nitrogen or air.

For instruments used with gases containing silicon or germanium compounds :

a. A solution of 2 % HF : 90% H_2O and 8% HNO_3 (3 to 5 minutes).
b. Distilled water (2 to 3 minutes)
c. Alcohol (2 to 3 minutes)
d. Blow dry with dry nitrogen or air.

Individual flow-path elements should be cleaned as follows:

Fittings

Remove the fittings from the base and clean individually.

Sensor

Remove the P.C.Board from the assembly, then remove the two screws which secure the sensor assembly to the base. Run a .007 inch diameter wire (stainless steel or piano) through the full length of the sensor tube 3 or 4 times. Replace the sensor on the base and flush with solvent during the Bypass Cleaning Procedure as described below.

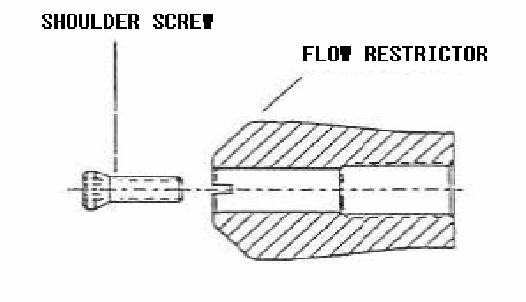


Figure 5
FC-280A/FC-280SA/FM-380A Bypass

Bypass FC-280A/FC-280SA/FM-380A (see Figure 5)

Loosen center locking shoulder screw one turn using a 7/64 inch Allen wrench. With a 1/4 inch flat head screw-drive, turn the bypass clockwise fully into the bore of the base. IMPORTANT: Count the exact number of turns it takes to hit bottom(N) and do not over-torque once the stop is reached. At this point, the bypass is closed and the sensor passageway can be solvent flushed and nitrogen purged through the inlet and outlet ports of the base. (Note: The valve will have to be mechanically opened or removed from the base during the operation.)

To clean the bypass, back it out of the bore by turning the bypass counter-clockwise. Either flush in place or, in the case of severe contamination build-up, remove and clean separately by screwing the bypass fully off of the shaft.

After cleaning, reinstall the bypass as follows:

1. Screw the bypass fully into the bore (clockwise) until it bottoms out in the bore. Do not over-torque once the stop is reached.
2. Turn the bypass exactly (N) turns counter-clock-wise (as previously noted during disassembly) thereby moving the bypass back out to its original axial position.
3. Tighten the shoulder screw into the threaded shaft with a torque of 15 to 20 inch-pounds. CAUTION! Do not over-torque.

If this operation is performed carefully, no further bypass adjustment should be required to re-establish the original calibration within the adjustment range of the potentiometer in the electronics.

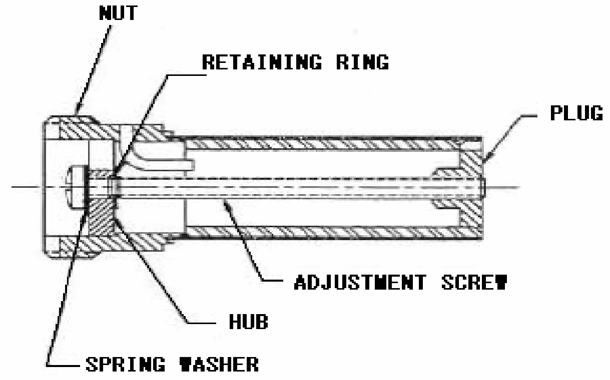


Figure 6
FC-282S Bypass

Bypass FC-282S (see Figure 6)

Remove the inlet fitting. Using a 3/4-inch wide flat blade screwdriver, turn bypass assembly counter clockwise until it becomes loose and can be removed. Note: Be careful not to turn the center adjustment screw. After cleaning, thread the bypass back into the flow body and firmly tighten against shoulder.

Valve FC-280A/FC-280SA (see Figure 7)

For minor contamination, the valve can be cleaned in place by mechanically opening the valve, flushing with solvent, purging with dry nitrogen and then resetting the valve during test as described in the Valve Adjustment Procedure. To mechanically open the valve, turn the adjusting nut on top of the valve 2 to 3 turns counter clockwise with a 5/16 inch open-end box wrench.

For severe contamination, the valve assembly and valve seat can be removed from the vase and cleaned separately. This is done as follows:

1. Unplug the valve from the P.C.Board(CAUTION! Always make sure power is off before disconnecting or reconnecting the valve to the P.C.Board). Remove(or loosen) the P.C.Board from the base assembly to provide clearance for the removal of the valve assembly.
2. Remove the two screw which secure the valve flange to the base and pull the valve assembly out completely.
3. If further disassembly is necessary (due to the level of contamination), remove the plunger assembly (plunger, poppet, and spring) by holding the inner sleeve assembly at the top of the valve with a 5/16 inch ring-wrench and removing the valve retaining nut with a 1/2 inch open-end wrench.
4. Thoroughly clean the plunger assembly and the inside of the inner sleeve assembly. Examine the poppet sealing surface and the valve O-ring and replace if necessary. (NOTE: Be sure to re-lubricate the O-ring with Halocarbon grease for Viton O-rings or Krytox grease for Kalrez O-rings prior to re-installation into the base).
5. Reinstall the plunger assembly making certain that the spider spring locates concentrically in the counter-bore of the inner sleeve assembly. Tighten the retaining nut with a torque of 20 (± 3) inch pounds and verify that the spring is positively held in place with no axial or rotational movement.
6. Replace the valve assembly by inserting it straight down into the base. Locate the valve coil leads towards the P.C.Board to provide maximum clearance for the valve adjustment wrench. Secure the valve in place by alternate tightening of the valve flange screws with 15 inch-pounds of torque.
7. Adjust the valve during test as described in the Valve Adjustment Procedure.

Valve FC-280S

Due to the complexity of the high flow valve with its necessary sequence of internal assembly, mechanical adjustments and/or disassembly should not be attempted by other than qualified Tylan General service personnel. To clean, do not disassemble the instrument. Energize the valve, solvent flush then purge with dry nitrogen for 1 hour.

After cleaning or replacement of the sensor, bypass or valve assemblies, the instrument must be adjusted and recalibrated against a suitable reference standard or calibrator to re-establish the original calibration and performance characteristics.

Adjustment and Calibration Procedures

In order to maintain the original factory calibration and performance characteristics of the instrument, the following adjustments may be necessary during routine maintenance and servicing and will definitely be required after disassembly for cleaning and/or parts replacement.

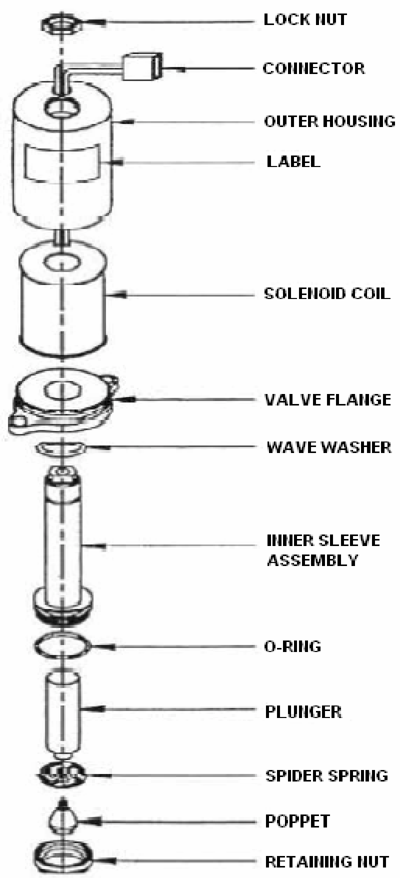


Figure 7
FC-280A/FC-280SA Solenoid Valve
904149-xxx, Rev.AD

Calibration - Indicated Versus Actual Flow Rate

Each controller is factory calibrated for the specific flow and gases indicated on the nameplate. Standard factory calibration is within ±1.0%. The calibration for other gases can be approximated to ±5% by using the conversion factors shown in the Conversion Factor Tables.

Factory adjustments should not be altered unless precision gas flow measuring equipment is available for calibration. Rotameters do not have sufficient accuracy for flow measurement calibration unless they have been specifically calibrated and the proper corrections are made for temperature and pressure. For access to the electronics, remove the two cover screws and carefully lift the cover from the assembly. If electrical adjustments are made with the cover removed and the PC board exposed, be very careful not to break wires or create short circuits while the instrument is open.

Recalibration FC-280SA/FC-280S

This may be accomplished as follows :

1. Thoroughly flush and dry the instrument to remove contaminants (see Cleaning Procedures).
2. Connect a source of gas to the inlet and a suitable flow standard to the outlet. If a volumetric calibrator is used, be sure to apply the proper density corrections to maintain the mass flow calibration.
3. Connect the power and indicator wires (see Figure 10A Electrical Hook-up Diagram).
4. Either mechanically open the valve by turning the adjusting nut on top of the valve 2 to 3 turns counter-clockwise, or electrically drive the valve open by connecting pin E to pin C (0 VDC). This will override the control loop, allowing flowmeter operation.
5. Remove the outlet gas line and cap off the instrument to assure zero flow through the sensor. Adjust ZERO potentiometer P1 to make the indicator read zero.
6. Reconnect the outlet gas line and adjust the flow to the full-scale value. Set the output to read 5.00 VDC at the full-scale flow rate using the Gain potentiometer P2.
7. Recheck zero as described in step 5.
8. Linearity adjustments are not normally required. After achieving calibration at zero and maximum flow, the midpoint calibration may be checked by setting the flow to one-half of full-scale. If the indicator does not read 2.50 ±0.025 VDC, adjustment the output to 2.50 VDC using the LINEARITY potentiometer P3. Although this adjustment is essentially independent, steps 5 and 6 should be repeated until all three points are within the desired calibration.
9. The flowmeter section of the flow controllers can be calibrated with the instrument operation in the controller mode and using the setpoint control to dial in the desired flow rate. Adjustments should be made as in steps 5 through 8. When adjusting the gain and linearity potentiometers, the actual flow will change rather than the output voltage to the commanded setpoint voltage.
10. If the instrument cannot be brought into calibration within the

adjustment range of P2 or P3, the bypass will have to be adjusted. This procedure is described in the Range Change section of this manual.

11. When using a test gas other than the intended usage gas, a correction factor equaling the ratio of the conversion factors of the two gases must be applied. See the Conversion Factors section, Appendix A for further explanation.

Dynamic Response Adjustment(FC-282SA/FC-280S only) potentiometer P4 provides the means for tuning the speed-up circuit, and R29 is to be changed for optimizing stability and dynamic response characteristics of the control loop. While these adjustments are factory set to give repeatable response characteristics independent of gas density and inlet pressure, the user may want to alter the settings for optimized performance under the particular operating conditions. This optimizing may also be required after range changes or after range changes or after replacement of sensor or control valve. When doing any readjustment the following procedure should be followed :

1. The first step is to verify the calibration of the mass flow controller. The procedure for this is detailed in the calibration section of this manual.
2. Start with the previously set adjustment or alternately center potentiometer P4, and set R29 TO 402kΩ

   NOTE : A resistance decade box is useful for finding the optimum value of R29.
3. Supply the instrument with the intended usage gas or a suitable substitute gas at the intended operating conditions including inlet pressure.
4. Generate step changes of the setpoint signal. For example, 50% to 100% and vice versa, and watch the output signal using a storage oscilloscope or a fast chart recorder.
5. Adjust potentiometer P4 and resistor R29 subsequently, until the instrument shows optimum response characteristics.

   NOTE : Speeding-up is accomplished by adjusting P4 clockwise. If instability occurs before the desired speed is established, adjust P4 counter clockwise until stability is achieved. Too low resistance values of R29 results in oscillation.

Recalibration FC-280A/FM-380S

This may be accomplished as follows :

1. Thoroughly flush and dry to remove contaminants(see Cleaning Procedures).
2. Connect a source of gas to the inlet and a suitable flow standard to the outlet. If a volumetric calibrator is used, be sure to apply the proper density corrections to maintain the mass flow calibration.
3. Connect the power and indicator wires (see Figure 10 Electrical Hook-up Diagram).
4. Either mechanically open the valve by turning the adjusting nut on top of the valve 2 to 3 turns counter-clockwise, or electrically drive

the valve open by connecting pin K to pin F (-15 VDC). This will override the control loop allowing flowmeter operation.

5. Remove the outlet gas line and cap-off the instrument to assure zero flow through the sensor. Adjust zero potentiometer R3 to make the indicator read zero.
6. Reconnect the outlet gas line and adjust the flow to the full-scale value. Set the output to read 5.00 VDC at the full-scale flow rate using the GAIN potentiometer R9.
7. Recheck zero as described in step 5.
8. Linearity adjustments are not normally required. After achieving calibration at zero and maximum flow, the midpoint calibration may be checked by setting the flow to make the indicator read 2.50 VDC. The calibrator should then measure half of the full-range flow rate within ± 0.5% of full-scale. If not, set the flow rate at one-half of the full-scale value, and adjust the output to 2.50 VDC using the LINEARITY potentiometer R19. Although this adjustment is essentially independent of the zero and full-scale adjustments, steps 5 and 6 should be repeated until all three points are within the desired calibration.
9. The flowmeter section of the flow controllers can be calibrated with the instrument operating in the controller mode and using the setpoint control to dial in the desired flow rate. Adjustments should then be made as in steps 5 through 8. When adjusting the gain and linearity potentiometers, the actual flow will change rather than the output voltage since the controller acts to control the output voltage to the commanded setpoint voltage.
10. If the instrument cannot be brought into calibration within the adjustment range of R9, the bypass will have to be adjusted. This procedure is described in the Range Change section of this manual.
11. If it is desired to calibrate the flow metering section for a gas other than that of the original factory calibration, determine the gas conversion factor relative to N2 from the Conversion Factor Tables and then activate the appropriate positions (1 through 7) of Dip Switch SW1 in accordance with Figure 15.

    NOTE : Factory calibration is done with the dip switch set to the conversion factor relative to N2 for the nameplate gas. If the conversion factor of the nameplate gas is less than 0.50, the dip switch is set to 1.00
12. When using a test gas other than the intended usage gas, a correction factor equaling the ratio of the conversion factors of the two gases must be applied. See the Conversion Factors section for further explanation.

Dynamic Response Adjustment (FC-280A only)

Dip Switch SW1, positions 10.11. and 12 provide the means for optimizing the stability and dynamic response characteristics of the control loop. While these switch positions are factory set to give repeatable response characteristics independent of gas density and inlet pressure, the user may want to alter the settings for optimized performance under his particular set of conditions (operation pressure, gas, and flow rates). To assist in doing so, the following explanation of these switch functions are presented (refer to Figure 14, FC-280A Schematic).

1. SW1 positions 10 and 11 provide four separate selections of the dynamic feedback resistance of the control loop. The higher the value of this resistance (positions 10 and 11 "off"), the higher the A.C. gain. This produces the best response and overshoot characteristics, but is prone to control instability. Switching positions 10 and / or 11 "on" reduces the resistance, thereby improving the stability at the expense of optinal response characteristics.
2. SW1 position 12 provides selection of the setpoint ramp rate from 1/2 second (first order time constant) in the "on" position, to 1 second in the "off" position. The longer time constant typically gives better over and undershoot characteristics in response to a step change in setpoint and provides a smoother transition in actual flow rate. The shorter time constant may be desirable when process runs are very short and the time to reach setpoint is critical to the process.
3. The optimal switch settings for full-scale flow rates of 20 SCCM and below are positions 10 and 11 "on" and position 12 is "off". For full-scale rates above 200 SCCM, best results have been achieved with positions 10 and 11 "off" and position 12 "on".
4. It should be noted that the required valve control voltage has some influence on the optimization of the dynamic response in that the higher voltages require a decrease in the dynamic feedback resistance to achieve stability (typically from 100K to 30K - i.e., SW1 position 11 "on").

Valve Adjustment (FC-280A/FC-280SA)

Valve Adjustment is accomplished through the following procedure :

1. Plumb the inlet of the instrument to a regulated supply of the appropriate gas. Connect a reference flowmeter in series, or monitor the flow as measured by the flow metering section of the instrument it self.
2. Remove the cover to permit access to the valve assembly. Plug the valve into the P.C.board prior to applying power to the instrument. Starting with the valve mechanically open and the input command signal at zero, slowly apply inlet pressure to the controller (to 40 psig). Mechanically close the valve by turning the adjusting nut clockwise until the flow is reduced to less than 2% of full scale.
3. Set the inlet pressure to the minimum (10 psig for up to 5 SLM full scale, and 15 psig for 6 to 30 SLM full scale) and command 100% flow rate. Verify full-scale output. On the FC-280A/FM-380A, set SW1 position 9 to "off" for full-scale flow rates of 1 SLM or less or "on" for full-scale flow rates greater than 1 SLM before setting the inlet pressure.
4. While monitoring the voltage applied to the valve (pin E to pin F), mechanically "fine-tune" the valve adjustment to achieve as close to the optimal 8 VDC control voltage as possible. Repeat this process

until the valve controls from <2% to 100 % full scale at both extremes of pressure.

5. After the valve is properly adjusted and tuned dynamically (see below), tighten the lock nut on the top of the valve with 15 inch-pounds of torque. This will lock the adjusted position in place.

Bypass Adjustment

If, during calibration, the desired calibration cannot be achieved within the range of the potentiometers, the bypass will have to be readjusted. The procedure for this is detailed in the Bypass Adjustment Procedure listed in the Range Change section below.

Range Change

Following are the procedures necessary to convert any instrument from one full scale flow range to another.

Sensor :

In order to change the range of a reactive gas instrument containing a large bore sensor (902343-009) to a full-scale flow higher than 3 slm, the sensor must be replaced with the small bore sensor (902343-008). This is done as follows :

1. Replace the sensor as detailed in the troubleshooting section.
2. FC-280SA/FC-282S only

   Install R37 (=3.48kΩ) for use with large bore sensor. Remove R37 for use with small bore sensor.

   FC-280A/FM-380A only

   Set Dip Switch SW1 position 8 to the proper sensor current (i.e., "off" for the small bore sensor to provide 13 milliamps or "on" for the large bore sensor to provide 15 milliamps).

   NOTE : Both steps are ultimately important since the increased current for the large bore sensor is necessary to compensate for additional heat loss to maintain the proper temperature rise, and is essential to achieve the specified accuracy and linearity of the flow signal.
3. Readjust the bypass as described in the Bypass Adjustment Procedure and recalibrate per the Calibration Procedure previously described.

Bypass FC-280A/FC-280SA/FC-380A :

The FC-280A, FC-280SA and FM-380A have been purposely designed to minimize the need to replace parts in order to change the full scale flow range of the instrument. The continuously adjustable bypass is the key to the wide range of the instrument. For conversion factors of less than 0.50 on the FC-280A/FM-380A only, set Dip Switch S1 positions 2,5, and 6 "off"(conversion factor equal to 1.00) prior to adjustment of the bypass. Full scale flow ranges of 10 SCCM to 30 SLM are easily achievable as follows :

1. Energize the instrument and adjust the flow rate to give 100%

indicated output. Center the gain pot of the electronics by rotating the adjustment screw 12 turns from its point of non-influence on either the indicated output (flowmeter) or the actual flow rate (flow controller).

2. Measure the actual flow rate (w1) at 100% indicated flow rate using a suitable reference flow standard.
3. Remove the inlet fitting and loosen the bypass as explained in the Cleaning Section.
4. Loosen the center locking screw one turn and turn the entire bypass clockwise. For reference, count the exact number of turns it takes to solidly bottom out. Do not torque.
5. Refer to the graph in Figure 8 for the number of turns from full insertion.
6. Turn the adjusting screw N turns clockwise, backing the bypass out to its new location.
7. Measure the actual flow rate at 100% indicated flow rate and if within ± 15% of the desired flow rate, adjust the gain potentiometer to bring it into calibration. If the actual flow is more than 15% different from the flow rate, refer to Figure 8 and readjust the bypass accordingly.
8. An in-line adjustment tool (P/N T-900-454) is available to adjust the bypass with gas flowing through toe instrument in lieu of steps 4 through7. Contact Tylan General for price and delivery.

Bypass FC-282S :

To change the range of the instrument, an adjustment may be made if the desired range is within the range of the installed bypass. Otherwise, it must be replaced. A three digit number etched on the bypass is used to identify its range as follows :

-001	25 to 75 slm
-002	76 to 150 slm
-003	151 to 300 slm

If the desired range is within the range of the installed bypass assembly, only a simple adjustment is necessary.

1. First, remove the bypass assembly as described in the cleaning section.
2. Using a Phillips head screwdriver, adjust the center screw by turning counter-clockwise to locate the plug flush with the down-stream end of the screen.
3. Thread the bypass back into the flow body and firmly tighten against shoulder.
4. Adjust the bypass by turning the adjusting screw clockwise to decrease the full scale flow rate.

Adjustment resolution for various high flow bypass sizes are approxi-mately as follows :

-001	1.2 slm per turn
-002	2.5 slm per turn
-003	5.0 slm per turn

CAUTION : Press the screw firmly against the bypass nut during adjustment to insure that the threads only engage the plug and not the hub. Make sure that the screw does not thread out of the assembly during counter-clockwise adjustment.

Valve FC-280A/FC-280SA :

Five valve seats with different orifice sizes are available to accommodate full scale ranges of 10 SCCM through 30 SLM. See Figure 17 for the flow ranges of each orifice. All five orifices are available in the Tylan General Spare Parts Kit.

When changing ranges, the valve seat may have to be changed accordingly. This can be accomplished by removing the valve assembly and replacing the valve seat as described in the Cleaning section. Readjustment of the valve is recommended (and in most cases is necessary) for optimal performance following any range change of 2:1 or more. Refer to the Valve Adjustment Procedure previously described.

Valve FC-282S

The FC-282S valve has two adjustments which allow the user to optimize its performance for the actual operating pressure. These adjustments can be made while the flow controller is operating and without removing the valve assembly from the base.

To adjust the minimum flow (leak-through), set the inlet pressure to 50 psig or the maximum expected and command zero flow. Loosen the lower knurled nut and rotate the valve housing clockwise to reduce the flow to between 1% and 3% of full scale. Retighten the knurled nut to lock the valve in place.

To adjust the full scale flow, set the inlet pressure to 20 psig or the minimum expected and command 100 % of range. Loosen the top lock nut and while monitoring the valve voltage (Pin E to Pin F) turn the top adjustment to give 100% flow at a valve voltage of 9 to 10 VDC. This will allow optimum response performance and provide adequate voltage margin to accommodate changes in operating conditions. Verify operation at 4%, 50% and 100% of the full scale range before and after tightening the lock nut.

After adjustments are made, it may be necessary to optimize dynamic response, refer to Dynamic Response Adjustment(FC-280SA/FC-282S) procedure.

Electronics FC-280SA/FC-282S :

The following components need to be adjusted or selected during the calibration procedure or dynamic response optimizing as previously described :

Potentiometer P1 Zero Adjustment

Potentiometer P2 Gain Adjustment

Potentiometer P3	Linearity Adjustment
Potentiometer P4	Dynamic response Adjustment
Resistor R29	Proportional band adjustment
Resistor R37	Sensor current Selection

Additionally, resistor R36 is be re-selected in case of replacement of the reference diode D5. The following value should be chosen as a function of the reference voltage (to be accessed through pin 6 of the card edge connector, labeled "Zener Test Point" , and measured against pin C, labeled "Return").

Voltage	Resistance (kΩ)
7. 2 VDC	14.3
7.1 VDC	10.5
7.0 VDC	8.25
6.9 VDC	6.81
6.8 VDC	5.62
6.7 VDC	4.87
6.6 VDC	4.22

Note : After replacement of the reference diode D5 or the quad operational amplifier IC1, recalibration of the instrument is required.

Electronics FC-280A/FM-380A :

Only those changes concerning the Dip Switch (SW1) as previously described need be made during a range change. To reiterate, positions 1 through 7 accommodate different gases by conversion factor selection, position 8 is used when converting from the large bore sensor to the small bore sensor or vice-versa. Position 9 allows for low differential pressure and/or higher flow applications. Positions 10 and 11 provide optimization of the response characteristics via the dynamic feedback resistor in the control loop. Position 12 allows selection of the setpoint ramp-rate-time constant.

Integral Display (FC-280AD/FM-380AD)

The numeric display is factory set for the nameplate range in engineering units (sccm or slm) The display may also be set to read as a percentage of full scale. If it becomes necessary to re-range the display, it is recommended that this service be performed by a factory authorized service center or contact Tylan General Technical Support for information.

TROUBLESHOOTING PROCEDURES

Initial Test

1. Check setup and procedure against the connection instructions given in the installation section. Permanent damage to the instrument may result if purging procedures are not followed, or if line power is accidentally applied to the signal leads.
2. Test line cord for compliance with pin assignments and continuity from all wires to correct pins (see Figures 10 and 10A Electrical Hook-up Diagram). Use a hipot tester to check for any pin-to-pin

shorts; during the test, flex the cable coming out of the connector to find intermittent shorts.

3. Check insulation resistance form pin 2 to base. It should exceed 50 megohms at 50 VDC. Pin 1 to case should measure less than 1 ohm.
4. Connect a source of gas (same as that for which the instrument is calibrated or equivalent test gas) to the inlet fitting. Apply power and allow a 10 minute warm-up period. With the inlet pressure between the specified range for the full scale flow rate, the output signal of the controller should follow the command setting (i.e.,50% of full scale at 2.50 VDC commanded voltage). The actual flow (as measured by a suitable flow standard in series with the controller) should follow the command setting and agree with the indicated flow of the controller. If the instrument is malfunctioning or cannot be recalibrated as previously described in the Adjustment and Calibration Procedure, check the appropriate symptom in the Troubleshooting Chart.

Tylan General has maintenance kits available which contain custom-designed tools that simplify the repair of all Tylan General's mass flow equipment.

REPLACEABLE SPARE PARTS

Description	Part Number
O-Ring Kit :	
(contains all O-Rings to rebuild a flow controller)	
FC-280A/FC-280SA/FM-380A:	
◈ Viton	907971-001
◈ Kalrez	907971-002
◈ Neporene	907971-003
Orifice Kit (contains all 5 orifices)	905491-001
(FC-280A/FC-280SA) :	
Sensor :	
◈ Viton	902343-008
◈ Kalrez (small bore)	902343-010
◈ Neporene	902343-012
Valve :	
FC-280A/FC-280SA	
◈ Viton	904149-001
◈ Kalrez	904149-003
◈ Neporene	904149-005
P.C.Board Assembly	
FC-280SA/FC-380SA	908210-001

GENERAL TROUBLESHOOTING

Symptom	Possible Cause	Remedy
No output	Faulty meter	Read output at Pins 3 and 2 directly with alternate tmeter.
	No actual flow	Check pressures, valve positions, line or filter blockage.
	Sensor clogged	See Cleaning Procedures.
	Valve closed	See Valve section. Measure valve DC resistance (should be 220 ohms ±10 ohms). If coil is open, replace coil or entire valve assembly.
	Electronic failure	See Electronics Section.
	No input power	Check for line power between appropriate pins at mating connector.
	Faulty power supply	Check input/output voltages (+ & −15 VDC, +5.00 VDC).
Maximum signal (approximately 200% of full scale): a: Indication correct, flow is high	Valve failed open	Check valve voltage as measured across valve wires. Valve should close when voltage decreases below 4.00 VDC. Higher voltage indicates lack of closing command or electronic failure; repair or replace electronics.
b: Indication erroneous	Faulty power supply or command signal	Check input/output (+ & −15 VDC, +5.00 VDC).
	Open resistance element on sensor	Replace sensor.
	Electronics failure	See Electronics Section.

ELECTRONICS TROUBLESHOOTING

(See Figure 10 and 10A Electrical Schematic and Note 1 through 3 list below)

Symptom	Possible Cause	Remedy
General failure or miscalibration	Power supply voltage off of nominal.	Check + & −15 VDC. Check sensor and valve or connections.
Flow indication saturated (−0.7 or +12.0 VDC) regardless of flow	Bridge or sensor failure.	Check sensor resistances, these should be 330 ohms ±10 ohms) green to red and green to brown with power off. Voltage across brown to circuit common should be 8.0 to 10.0 VDC and Ez to circuit common should be −6.9 ±0.3 VDC.
	Component failure.	Check zero pot, gain pot, other components and solder joints.
Valve drive open or saturated(FC-280SA only) (0 or 12 to 15 VDC)	Valve drive transistor open or shorted. Op-amp failed.	Check these and other components and solder joints. Replace if necessary.
All circuits functional, but out of calibration	Contamination, or as a result of cleaning or repairing.	Adjust.
Instrument Controls, but output voltage does not agree with control input pot voltage	+5.00 VDC supply of the control input off nominal.	Check +5.00 VDC input reference supply. Readjust as necessary.
	Large input voltage offset in Op-amp	Replace if necessary, or replace PC Board assembly.

VALVE TROUBLESHOOTING

Symptom	Possible Cause	Remedy
Signal offset at zero flow	Electronics not adjusted.	See Adjustment Procedures.
Valve fails to close	Contamination.	See Cleaning Procedures.
	Electronics failure.	See Electronics Section.
	Mechanical damage from over-pressure or other cause.	Adjust or replace valve.
	Operation on wrong gas (e.g., instrument adjusted for argon may not shut off completely on hydrogen).	Test on proper gas or mechanically readjust valve.
Valve fails to open	Contamination.	See Cleaning Procedures.
	Electrically commanded closed or pot shorted	Check command signal ((Pins A and B) and pot. Check for Electronic failure.
Valve controls at flow rates, but not at minimum	Contamination.	See Cleaning Procedures.
	Erosion or corrosion.	Replace valve.
	Improper adjustment or inadequate drive.	See Adjustment Procedures.
Valve oscillates or hunts	Improper system dynamics due to excessive inlet pressure.	Reduce upstream pressure regulator setting..
	Improper dynamics in electronics.	Check for failed resistor or capacitor. See Dynamic Response Adjustment Procedure.

NOTE 1 : CAUTION: Always command zero flow when the gas supply is shut off. Failure to do so will result in the valve overheating and in excessive flow transients after the gas supply is turned “on”(see Figures 10 and 10A Electrical Hook-Up Diagram).

NOTE 2: Pin D may be connected to common prior to turning on the gas supply to achieve “soft-start” performance.

NOTE 3 : A threshold circuit in the controller ensures zero valve voltage whenever the commanded signal falls below 1.0% full scale.

Sensor Replacement

If it is determined that either of the sensor elements is open or shorted to case, it will be necessary to replace sensor assembly. This can be done by unsoldering the three sensor leads from the P.C.Board and and removing the three screws which attach the sensor assembly to the base and the P.C.Board. The sensor assembly can then be removed and replaced. Before installing the new sensor element, resistance should be checked and measured as follows:

Upstream...Ru=330±10 ohms
(Brown to Green)

Downstream..Rd=330±10 ohms
(Red to Green)

Isolation.. > 100 megohms
(All leads to sensor housing)

Solenoid Valve Replacement

If it is determined that the coil of the solenoid valve is open, shorted to case or outside the DC resistance range of 115 to ohms, the valve assembly will need to be replaced.

Figure 9 - Specifications

Specifications	Mass Flowmeters FM-380A / FM-380AD	Mass Flow Controllers FC-280SA/FC-280A/FC-280AD
PERFORMANCE		
Full-scale range* (N2 equivalent)	10 sccm to 30 slm	10 sccm to 30 slm
Turndown ratio	50 : 1	50 : 1
Accuracy	± 1.0% of full scale	± 1.0% of full scale
Repeatability	±0.2%	±0.2%
Linearity	±0.5%	±0.5%
Step Response Time(0 to 100%)	2 seconds	FC-280S : 400 to 800 ms FC-280A : 2 seconds
OPERATING		
Maximum Operating Pressure	500 psig	150 psig
Proof Pressure	1500 psig	1500 psig
Differential Pressure	0.5 psid maximum	10 – 40 psid (to 5 slm) 15-40 psid (6 to 30 slm)
Pressure Coefficient	±0.007%/psi(typical)	±0.007%/psi(typical)
Temperature Range	0 – 50 ℃	0 – 50 ℃
Temperature Coefficient	±0.1%/℃	±0.1%/℃
Warm-up Time	30 minutes	30 minutes
Attitude Sensitivity	< 0.25% @ 90°	< 0.25% @ 90°
ENVIRONMENTAL		
Pressure	> 100 Torr	> 100 Torr
Temperature Range	0 – 50 ℃	
Humidity	0-95% relative humidity, non-condensing	
ELECTRICAL		
Supply Voltage:	+15 VDC ±4% 35mA -15 VDC ±4% 35mA	+15 VDC ±4% 35mA -15 VDC ±4% 150mA
Power	1 watt max	3 watt max
Output Voltage	0 – 5 VDC (into 2000 Ohms, shortcircuit protected)	
Control Voltage	0 – 5 VDC (Controller only from voltage source with maximum of 5000 Ohms)	
MECHANICAL		
Valve(> 1 million cycles)	High Speed Electromagnetic Normally Closed Solenoid	
Materials	Type 316 & 446 stainless steel, PFA teflon	
Seals	Viton® , Kalrez® or Neporene®	
Leak Integrity	1 X 10^{-9} atm cc/sec Helium	
Valve Through-Leak(Teflon Poppet)	<2% Full scale	
Fittings Available	1/8",1/4",3/8" VCR TM or Swagelok TM	
DIMENSIONAL	See Figure 16	

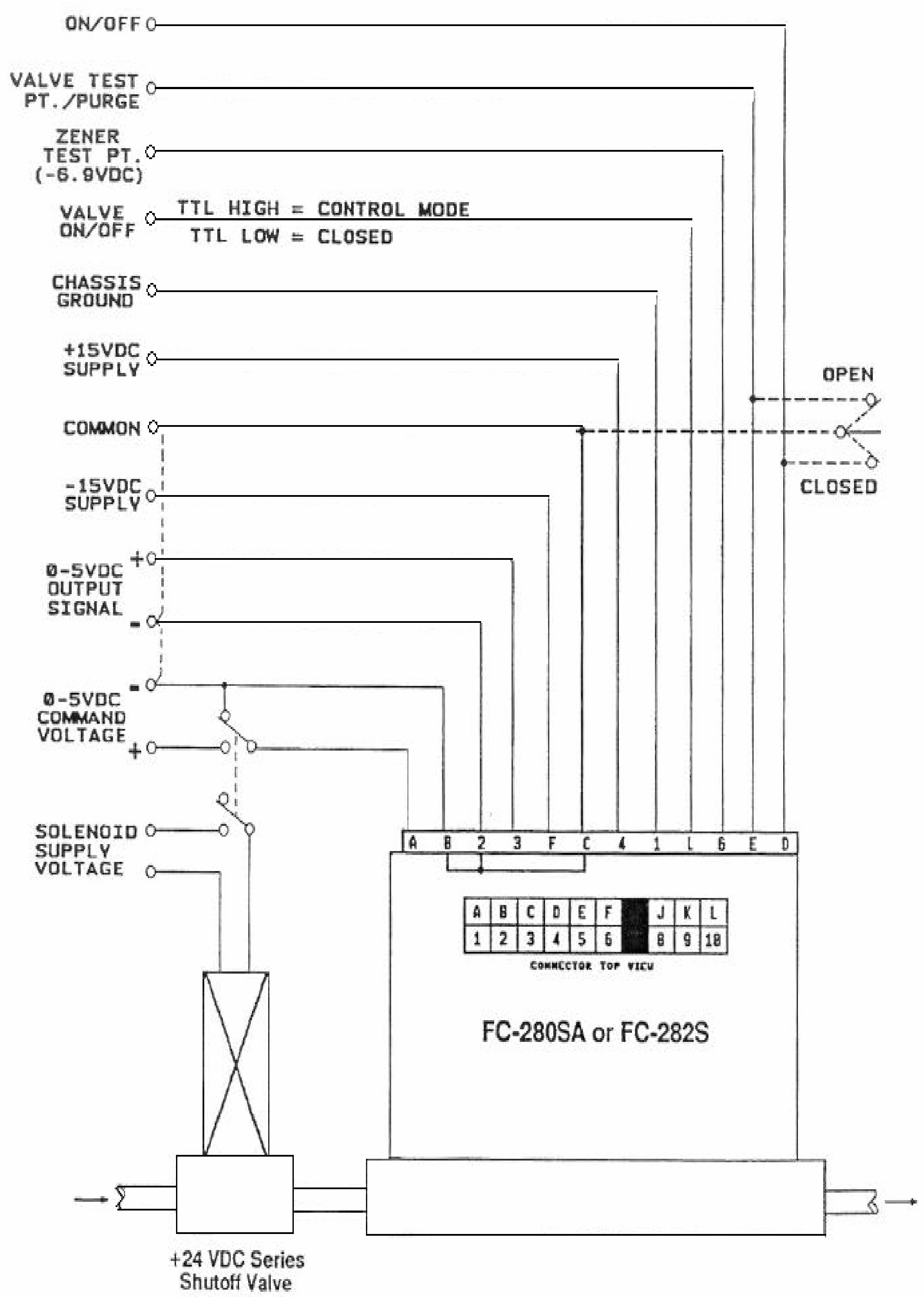
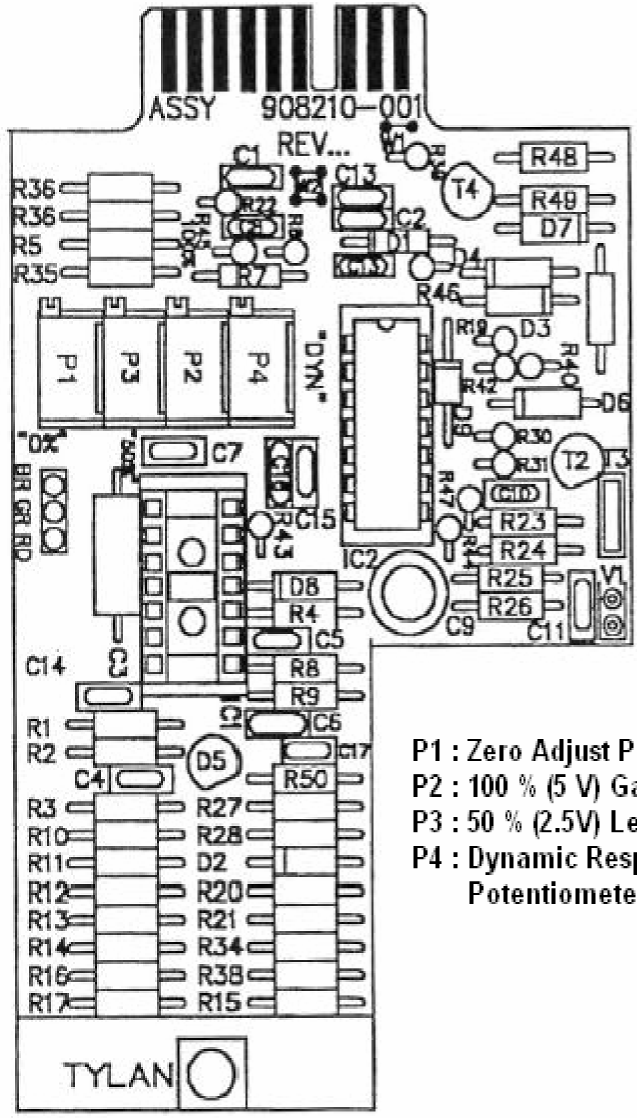


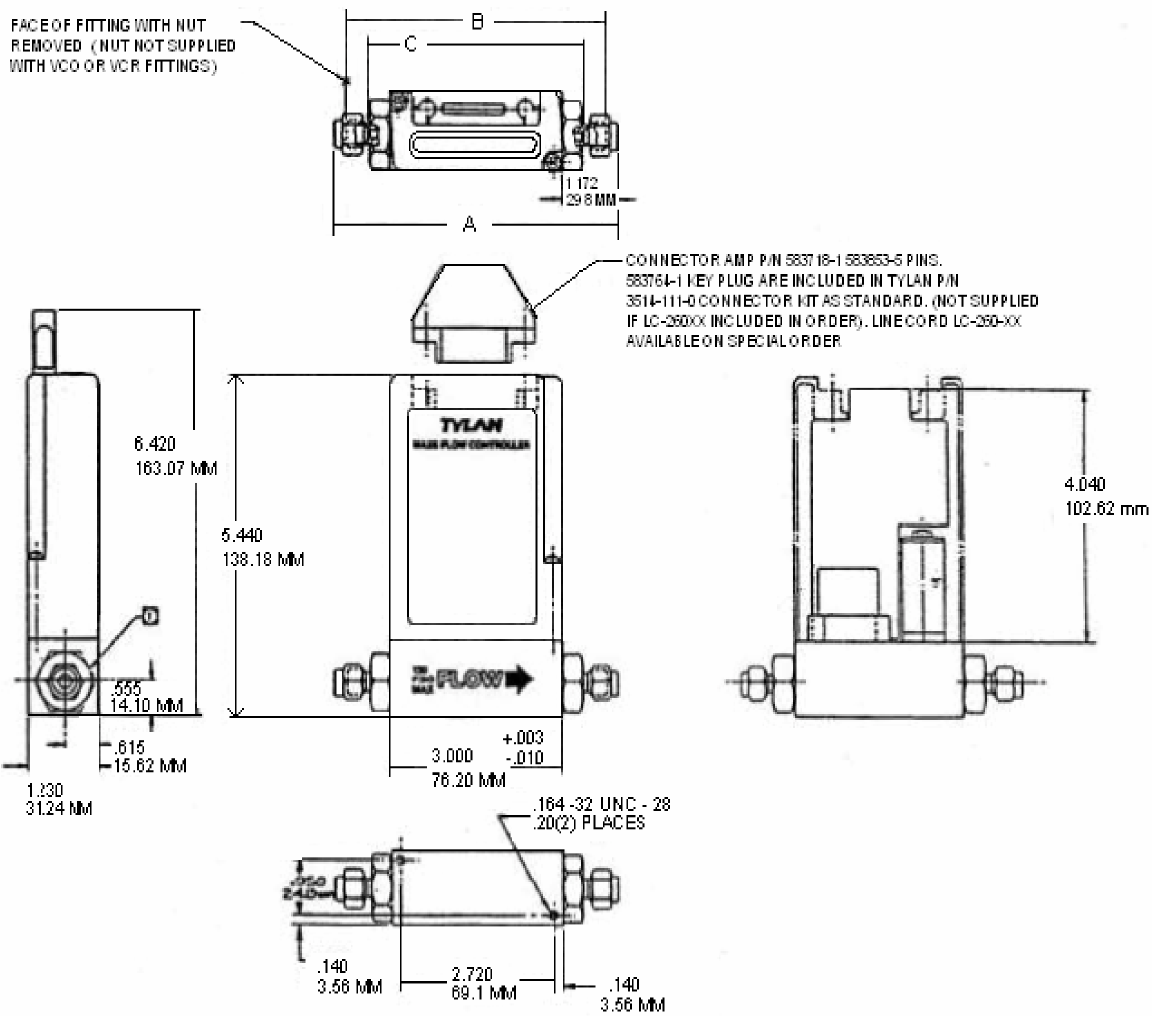
Figure 10A
FC-280SA / FC-282S Electrical Hook-Up Diagram



P1 : Zero Adjust Potentiometer
P2 : 100 % (5 V) Gain Potentiometer
P3 : 50 % (2.5V) Lenearity Potentiometer
P4 : Dynamic Response Adjust
Potentiometer

Figure 11A

908210001, Rev.T
P.C.B Assembly
FC-280SA Mass Flow Controller



INSTALLATION DIMENSIONS				
CODE	FITTING E SIZE	A DIMENSION	B DIM BASE + BODY LGTH	C DIM INSIDE FTG
2S	1/8 SWAGELOK	4.84 122.93 MM	4.32 109.72 MM	3.84 97.53 MM
4S	1/4 SWAGELOK	5.03 127.81 MM	4.44 112.77 MM	3.83 20.82 MM
4O	1/4 VCO	—	4.54 115.31 MM	—
4V	1/4 VCR	—	4.88 123.95 MM	—
6S	3/8 SWAGELOK	5.15 130.96 MM	4.56 115.82 MM	3.83 97.43 MM
6O	3/8 VCO	—	4.80 121.92 MM	—
6V	3/8 VCR	—	5.18 131.57 MM	—

Figure 16
FC-280A/FC-280SA/FM-380A
Mass Flow Controller/Flowmeter Outline

MASS FLOW CONVERSION FACTORS

Use of Gas Conversion Factors

The gas conversion factors is a dimensionless ratio describing the sensitivity of a mass flow controller to two different gases. Although this ratio is usually reported as a dimensionless number, the calculations can be followed much more easily if dimensions are added. The gas conversion factor for nitrogen is usually set to one, and then all other gases are referenced back to nitrogen. To determine the gas conversion factor referenced to a gas other than nitrogen, simply ratio the gas conversion factors of both gases referenced to nitrogen. The magnitude of the conversion factor is a function of the heat capacity, density, and molecular structure. A detailed description of how the gas conversion factors are derived is provided in section 3.0. Under normal conditions, it is not necessary for the customer to calculate a gas conversion factor from the heat capacity and gas density data.

This section contains a few examples to help understand how the conversion factors are applied. The conversion factors are primarily used when recalibrating a flow controller or flowing a gas other that the gas for which the controller was calibrated. The first example shows how to calibrate a mass flow controller with the test gas provided in the manual. In order to minimize the errors created in using conversion factors, the test gas is chosen so that the gas conversion factor is as close to one as possible. If the only gas available for calibration was something other than the test gas or nitrogen, then the gas conversion factor must be calculated. This process is shown in example 2. If you are using a mass flow controller for a gas other than the one for which it was calibrated, but you didn't want to recalibrate the MFC, then example 3 demonstrates how to determine the actual flow rate of the gas through the MFC.

Example 1 : Calibrate a 500 sccm Ammonia(NH_3) flow controller with Nitrous Oxide.

Conversion Factor (NH_3 to N_2O) = 1.014

Calculate the flow rate of Nitrous Oxide which equals 500 sccm and 250 sccm of Ammonia.

$$\frac{500 \text{ sccm } NH_3}{1.014\ NH_3 / N_2O} = 493 \text{ sccm } N_2O \qquad \frac{250 \text{ sccm } NH_3}{1.014\ NH_3 / N_2O} = 247 \text{ sccm } N_2O$$

Example 2 : Calibrate the same 500 sccm Ammonia flow controller with Argon.

Conversion Factor (NH_3 to N_2) = 0.719

Conversion Factor (Ar to N_2) = 1.415

$$\text{Conversion Factor } (NH_3 \text{ to } Ar) = \frac{0.719\ NH_3 / N_2}{1.415\ Ar / N_2} = 0.508\ NH_3 / Ar$$

Calculate the flow rate of Argon which equals 500 sccm and 250 sccm of Ammonia.

$$\frac{500 \text{ sccm } NH_3}{0.508\ NH_3/Ar} = 984 \text{ sccm Ar} \qquad \frac{250 \text{ sccm } NH_3}{0.508\ NH_3/Ar} = 492 \text{ sccm Ar}$$

Example 3 : 200 sccm Carbon Monoxide (CO) flow controller is to be temporarily used in a Carbon Dioxide (CO_2) application. How many sccm of CO_2 are flowing when the reading is 73% ?

Conversion Factor (CO to N_2) = 1.000

Conversion Factor (CO_2 to N_2) = 0.737

$$\text{Conversion Factor } (CO_2 \text{ to } CO) = \frac{0.737\ CO_2 / N_2}{1.000\ CO / N_2} = 0.737\ CO_2 / CO$$

(73%) X (200 sccm CO) X (0.737 CO_2 / CO) = 108 sccm CO_2

CONVERSION FACTOR TABLE (relative to nitrogen and test gas)

GAS	SYMBOL	Cal. Test Gas	CONVERSION FACTOR REL. TO Cal. Test Gas	N2	DYNAMIC RESP. TEST GAS	SPECIFIC Heat@25℃ Cp(cal/g℃)	DENSITY g/l@0℃	SUGG. SEAL MAT.	STD CAL PRESS (PSID)	OVERSIZE ORIFICE
Acetylene	C2H2	C2H4	0.973	0.581	C2H4	0.4049	1.1620	V	10 to 40	No
Air	AIR	N2	1.006	1.006	N2	0.2389	1.2930	V	10 to 40	No
Allene(Propadiene)	C2H4	C2H6	0.877	0.421	Ar	0.3633	1.7870	V	10 to 40	No
Ammonia	NH3	N20	1.014	0.719	N2	0.5005	0.7600	N	10 to 40	No
Argon	Ar	Ar	1.000	1.415	Ar	0.1245	1.7820	V	10 to 40	No
Arsine	AsH3	N20	0.949	0.673	N20	0.1168	3.4780	V	10 to 40	No
Boron Tribromide	BBr3	C2H6	0.787	0.378	SF6	0.0647	11.1800	KZ	3 to 12	Yes
Boron Trichloride	BCl2	C2H6	0.895	0.430	SF6	0.1217	5.2270	KZ	3 to 12	Yes
Boron Trifluoride	BF3	C2H6	1.057	0.508	N20	0.1779	3.0250	KZ	10 to 40	No
Bromine	Br2	N20	1.140	0.808	SF6	0.0539	7.1300	V	10 to 40	No
Bromine Pentafluoride	BrF5	SF6	0.596	0.253	SF6	0.1387	7.8030	KZ	3 to 12	Yes
Bromine Trifluoride	BrF3	C2H6	0.802	0.385	SF6	0.1162	6.1080	KZ	10 to 40	No
Bromotrifluoromethane(Freon-13B1)	CBrF3	SF6	1.399	0.370	SF6	0.1113	6.6440	V	3 to 12	Yes
1,3-Butadiene	C4H6	SF6	1.181	0.312	N20	0.3632	2.4130	V	6 to 24	Yes
n-Butane	n-C4H10	SF6	0.966	0.255	N20	0.4130	2.5930	V	6 to 24	Yes
1-Butane	C4H8	SF6	1.110	0.294	N20	0.3723	2.5030	V	10 to 40	No
2-Butane CIS	C4H8	SF6	1.070	0.283	N20	0.3863	2.5030	V	6 to 24	Yes
2-Butane TRANS	C4H8	SF6	1.187	0.314	N20	0.3483	2.5030	V	6 to 24	Yes
Carbon Dioxide	CO2	N20	1.040	0.737	N20	0.2017	1.9640	V	10 to 40	No
Carbon Disulfide	CS2	C2H4	1.003	0.600	N20	0.1434	3.3970	V	10 to 40	No
Carbon Monoxide	CO	N2	1.000	1.000	N2	0.2488	1.2500	V	10 to 40	No
Carbon Tetrachloride	CCl4	SF6	1.163	0.307	SF6	0.1297	6.8600	V	3 to 12	Yes
Carbon Tetrafluoride(Freon-14)	CF4	C2H6	0.874	0.420	N20	0.1659	3.9260	V	10 to 40	No
Carbonyl Fluoride	COF2	C2H4	0.908	0.542	N20	0.1712	2.9450	KZ	10 to 40	No
Carbonyl Sulfide	COS	N20	0.931	0.660	N20	0.1652	2.6800	KZ	10 to 40	No
Chlorine	Cl2	N20	1.210	0.858	N20	0.1145	3.1630	V	10 to 40	No
Chlorine Trifluoride	ClF3	C2H6	0.835	0.401	N20	0.1651	4.1250	KZ	3 to 12	Yes
Chlorodifluoromethane(Freon-22)	CHClF2	C2H6	0.934	0.449	N20	0.1580	3.8580	KZ	10 to 40	No
Chloroform	CHCl3	C2H6	0.816	0.392	SF6	0.1310	5.3260	V	3 to 12	Yes
Chloropentafluoroethane(Freon-115)	C2ClF5	SF6	0.915	0.242	SF6	0.1640	6.8959	V	10 to 40	No
Chlorotrifluoromenthane(Freon-13)	CClF3	SF6	1.243	0.329	SF6	0.1786	4.6600	V	10 to 40	No
Cyanogen	C2N2	C2H6	0.940	0.452	N20	0.2608	2.3220	V	10 to 40	No
Cyanogen Chloride	ClCN	C2H4	1.012	0.605	N20	0.1762	2.7420	V	10 to 40	No

GAS	SYMBOL	Cal. Test Gas	CONVERSION FACTOR REL. TO Cal. Test Gas	N2	DYNAMIC RESP. TEST GAS	SPECIFIC Heat@25℃ Cp(cal/g℃)	DENSITY g/l@0℃	SUGG. SEAL MAT.	STD CAL PRESS (PSID)	OVERSIZE ORIFICE
Cyclopropane	C3H6	C2H6	0.927	0.445	N2O	0.3272	1.8770	V	10 to 40	No
Deuterium	D2	N2	0.998	0.998	He	1.7325	0.1798	V	10 to 40	No
Diborane	B2H6	C2H6	0.918	0.441	N2	0.5020	1.2350	V	10 to 40	No
Dibromodiflouromethane	CBr2F2	SF6	1.210	0.320	SF6	0.0913	9.3620	V	3 to 12	Yes
Dibromomethane	CH2Br2	C2H6	0.978	0.470	SF6	0.0750	7.7600	V	10 to 40	No
Dichlorodifluoromethane(Freon-12)	CCl2F2	SF6	1.339	0.354	SF6	0.1432	5.3950	V	10 to 40	No
Dichlorofluoromethane(Freon-21)	CHCl2F	C2H6	0.875	0.420	SF6	0.1417	4.5920	V	10 to 40	No
Dichloromethylsilane	(CH3)2SiCl2	SF6	0.995	0.252	SF6	0.1882	5.7580	KZ	3 to 12	Yes
Dichlorosilane	SiH2Cl2	C2H6	0.858	0.412	SF6	0.1472	4.5060	KZ	3 to 12	Yes
1,2-Dichlorotetrafluoroethane(Freon-114)	C2Cl2F4	SF6	0.858	0.227	SF6	0.1581	7.6260	V	10 to 40	No
1,1-Difluorosilane	C2H2F2	C2H6	0.887	0.426	N2O	0.2245	2.8570	V	10 to 40	No
Dimethylamine	(CH3)2NH	SF6	1.346	0.356	N2O	0.3822	2.0110	V	3 to 12	No
Dimethyl Cadmium	(CH3)2Cd	SF6	0.736	0.194	SF6	0.2211	6.3607			
Dimethyl Ether	(CH3)2O	C2H6	0.812	0.390	N2O	0.3411	2.0550	KZ	10 to 40	No
2,2-Dimethylpropane	C5H12	SF6	0.821	0.217	N2O	0.3916	3.2190	V	3 to 12	Yes
Disilane	Si2H6	SF6	1.200	0.317	N2O	0.3105	2.7760	V	10 to 40	No
Ethane	C2H6	C2H6	1.000	0.481	C2H6	0.4241	1.3420	V	10 to 40	No
Ethanol	C2H6O	C2H6	0.815	0.392	N2O	0.3398	2.0550	V	3 to 12	Yes
Ethyl Acetylene(1-Butyne)	C4H6	SF6	1.220	0.322	N2O	0.3515	2.4130	V	3 to 12	Yes
Ethylamine	C2H5NH2	SF6	1.335	0.353	N2O	0.3854	2.0110	V	10 to 40	No
Ethyl Chloride	C2H5Cl	SF6	1.475	0.390	N2O	0.2436	2.8790	V	3 to 12	Yes
Ethylene	C2H4	C2H4	1.000	0.598	C2H4	0.3658	1.2510	V	10 to 40	No
Ethylene Oxide	C2H4O	C2H6	1.114	0.535	N2O	0.2601	1.9650	KZ	3 to 12	Yes
Fluorine	F2	N2	0.931	0.931	Ar	0.1970	1.6950	V	10 to 40	No
Fluroform(Freon-23)	CHF3	C2H6	1.045	0.502	N2O	0.1742	3.1270	V	10 to 40	No
Freon-11	CClF3	SF6	1.163	0.307	SF6	0.1452	6.1290	V	3 to 12	Yes
Freon-12	CCl2F2	SF6	1.339	0.354	SF6	0.1432	5.3950	V	10 to 40	No
Freon-13	CClF3	SF6	1.243	0.329	SF6	0.1786	4.6600	V	10 to 40	No
Freon-13B1	CBrF3	SF6	1.339	0.370	SF6	0.1113	6.6440	V	10 to 40	No
Freon-14	CF4	C2H6	0.874	0.420	N2O	0.1659	3.9260	V	10 to 40	No
Freon-21	CHCl2F	C2H6	0.875	0.420	SF6	0.1417	4.5920	KZ	3 to 12	Yes
Freon-22	CHClF2	C2H6	0.934	0.449	N2O	0.1580	3.8580	KZ	10 to 40	No
Freon-113	CCl2FCClF2	SF6	0.858	0.227	SF6	0.1442	8.3600	V	3 to 12	Yes
Freon-114	C2Cl2F4	SF6	0.858	0.227	SF6	0.1581	7.6260	V	6 to 24	Yes
Freon-115	C2ClF5	SF6	0.915	0.242	SF6	0.1640	6.8959	V	10 to 40	No
Freon-C318	C4F6	SF6	0.610	0.161	SF6	0.2021	8.3970	V	3 to 12	Yes
Germane	GeH4	C2H4	0.953	0.569	N2O	0.1405	3.4180	V	10 to 40	No
Germanium Tetrachloride	GeCl4	SF6	1.009	0.267	SF6	0.1072	9.5650	V	3 to 12	Yes
Helium	He	He	1.000	1.415	He	1.2418	0.1786	V	10 to 40	No
Hexafluoroethane(Freon-116)	C2F6	SF6	0.942	0.241	SF6	0.1844	6.1570	KZ	10 to 40	No
n-Hexane	C6H14	SF6	0.678	0.179	N2O	0.3971	3.8450	V	3 to 12	Yes
Hydrogen	H2	H2	1.000	1.010	H2	3.4224	0.0899	V	10 to 40	No
Hydrogen Bromide	HBr	N2	1.000	1.000	N2O	0.0861	3.6100	KZ	10 to 40	No
Hydrogen Chloride	HCl	N2	1.000	1.000	Ar	0.1911	1.6270	KZ	10 to 40	No

GAS	SYMBOL	Cal. Test Gas	CONVERSION FACTOR REL. TO Cal. Test Gas	N2	DYNAMIC RESP. TEST GAS	SPECIFIC Heat@25℃ Cp(cal/g℃)	DENSITY g/l@0℃	SUGG. SEAL MAT.	STD CAL PRESS (PSID)	OVERSIZE ORIFICE
Hydrogen Cyanide	HCN	N2O	1.077	0.764	N2	0.3172	1.2060	KZ	10 to 40	No
Hydrogen Fluoride	HF	N2	1.000	1.000	N2	0.3482	0.8930	KZ	3 to 12	Yes
Hydrogen Iodide	HI	N2	0.999	0.999	SF6	0.0545	5.7070	KZ	10 to 40	No
Hydrogen Selenide	H2Se	N2O	1.112	0.788	N2O	0.1026	3.6130	KZ	10 to 40	No
Hydrogen Sulfide	H2S	N2O	1.190	0.844	Ar	0.2278	1.5200	KZ	10 to 40	No
Iodine Pentafluoride	IF5	SF6	0.943	0.249	SF6	0.1109	9.9000	KZ	3 to 12	Yes
Isobutane	CH(CH3)3	SF6	0.997	0.264	N2O	0.3998	2.5950	V	6 to 24	Yes
Isobutylene	C4H6	SF6	1.081	0.286	N2O	0.3824	2.5030	V	6 to 24	Yes
Isopropanol	C2H7OH	SF6	1.093	0.289	N2O	0.3529	2.6830	V	3 to 12	Yes
Krypton	Kr	Ar	1.000	1.415	N2O	0.0593	3.7390	V	10 to 40	No
Methane	CH4	N2O	1.014	0.719	N2	0.5318	0.7150	V	10 to 40	No
Methanol	CH3OH	C2H4	0.977	0.584	C2H6	0.3277	1.4290	V	10 to 40	No
Methylacetylene(Propyne)	C3H4	C2H6	0.867	0.417	Ar	0.3672	1.7870	V	10 to 40	No
Methylamine	CH3NH2	C2H6	1.065	0.512	C2H6	0.3856	1.3860	V	10 to 40	No
Methyl Bromide	CH3Br	C2H4	0.953	0.569	N2O	0.1134	4.2360	V	6 to 24	Yes
Methyl Chloride	CH3Cl	C2H4	1.053	0.629	N2O	0.1929	2.2530	V	10 to 40	No
Methyl Fluoride	CH3F	N2O	0.964	0.684	C2H6	0.2636	1.5180	V	10 to 40	No
Methyl Mercaptan	CH3SH	C2H6	1.044	0.502	N2O	0.2541	2.1460	V	6 to 24	Yes
Methyl Trichlorosilane	(CH3)SiCl3	SF6	0.947	0.250	SF6	0.1638	6.6690	KZ	3 to 12	Yes
Molybdenum Hexafluoride	MoF6	SF6	0.571	0.151	SF6	0.1933	9.3719	KZ	3 to 12	Yes
Neon	Ne	Ar	1.000	1.415	N2	0.2464	0.9000	V	10 to 40	No
Nitric Oxide	NO	N2	0.976	0.976	N2	0.2378	1.3390	V	10 to 40	No
Nitrogen	N2	N2	1.000	1.000	N2	0.2486	1.2500	V	10 to 40	No
Nitrogen Dioxide	NO2	N2O	1.044	0.741	N2O	0.1923	2.0520	KZ	10 to 40	No
Nitrogen Trifluoride	NF3	C2H6	0.999	0.480	N2O	0.1798	3.1680	V	10 to 40	No
Nitrogen Trioxide	N2O3	C2H6	0.813	0.391	N2O	0.2063	3.3930	V	10 to 40	No
Nitrosyl Chloride	NOCl	C2H4	1.027	0.614	N2O	0.1629	2.9200	V	10 to 40	No
Nitrous Oxide	N2O	N2O	1.000	0.709	N2O	0.2098	1.9640	V	10 to 40	No
Octafluorocyclobutane(Freon-C318)	C4F8	SF6	0.610	0.161	SF6	0.1899	8.9370	V	10 to 40	No
Oxygen	O2	N2	0.992	0.992	N2	0.2196	1.4270	V	10 to 40	No
Oxygen Difluoride	OF2	C2H4	1.058	0.632	N2O	0.1918	2.4090	KZ	10 to 40	No
Ozone	O3	N2O	0.984	0.698	N2O	0.1954	2.1428	V	10 to 40	No
Pentaborane	B5H9	SF6	1.034	0.273	N2O	0.3555	2.8160	V	3 to 12	Yes
n-Pentane	n-C5H12	SF6	0.807	0.213	N2O	0.3984	3.2190	V	3 to 12	Yes
Perchloryl Fluoride	ClO3F	C2H6	0.822	0.359	SF6	0.1515	4.5710	V	10 to 40	No
Perfluoropropane	C3F8	SF6	0.635	0.168	SF6	0.1944	8.3880	V	10 to 40	No
Phosgene	COCl2	C2H6	0.925	0.444	SF6	0.1393	4.4180	V	3 to 12	Yes
Phosphine	PH3	N2O	0.974	0.691	C2H6	0.2610	1.5170	V	10 to 40	No
Phosphorous Oxychloride	POCl3	SF6	1.142	0.302	SF6	0.1324	6.8452	V	3 to 12	Yes
Phosphorous Pentafluoride	PF5	SF6	1.143	0.302	SF6	0.1611	5.6200	KZ	10 to 40	No
Phosphorous Tribromide	PBr3	SF6	1.275	0.337	SF6	0.0671	12.0842	V	3 to 12	Yes
Phosphorous Trichloride	PCl3	SF6	1.355	0.358	SF6	0.1247	6.1270	V	3 to 12	Yes
Propane	C3H8	SF6	1.318	0.348	N2O	0.3991	1.9670	V	10 to 40	No
Propylene	C3H6	C2H6	0.829	0.398	N2O	0.3659	1.8770	V	10 to 40	No

GAS	SYMBOL	Cal. Test Gas	CONVERSION FACTOR REL. TO Cal. Test Gas	N2	DYNAMIC RESP. TEST GAS	SPECIFIC Heat@25℃ Cp(cal/g℃)	DENSITY g/l@0℃	SUGG. SEAL MAT.	STD CAL PRESS (PSID)	OVERSIZE ORIFICE
Silane	SiH4	C2H4	1.001	0.599	C2H4	0.3189	1.4330	V	10 to 40	No
Silicon Tetrachloride	SiCl4	SF6	1.074	0.284	SF6	0.1270	7.5847	KZ	3 to 12	Yes
Sulfuryl Fluoride	SO2F2	C2H6	0.808	0.388	SF6	0.1545	4.5620	V	3 to 12	Yes
Tetraethoxysilane(TEOS)	Si(OC2H5)4	SF6	0.295	0.078	SF6	0.3770	9.3004	KZ	1 to 5	Yes
Tetrafluorahydrazine	N2F4	SF6	1.222	0.324	SF6	0.1821	4.6400	V	3 to 12	Yes
1,3,5,7-Tetramethylcyclotetrasiloxane(TMCTS)	[-His(CH3)O-]4	SF6			SF6		10.7366	KZ	1 to 5	Yes
Titanium Tetrachloride	TiCl4	SF6	0.778	0.206	SF6	0.1572	8.4650	V	10 to 40	No
Trichlorethane(TCA)	C2H3Cl3	SF6	1.051	0.278	SF6	0.1654	5.9500	V	10 to 40	No
Trichloroethylene(TCE)	C2HCl3	SF6	1.210	0.320	SF6	0.1459	5.8620	V	10 to 40	No
Trichlorofluoromethane(Freon-11)	CCl3F	SF6	1.163	0.307	SF6	0.1452	6.1290	V	10 to 40	No
Trichlorosilane	SiHCl3	SF6	1.285	0.340	SF6	0.1332	6.0430	KZ	3 to 12	Yes
Trichloro-Trifluoroethane(Freon-113)	CCl2FCClF2	SF6	0.858	0.227	SF6	0.1442	8.3600	V	3 to 12	No
Triisobutyl Aluminum	(C4H9)3Al	SF6	0.230	0.061	SF6	0.5080	8.8480	V	3 to 12	Yes
Trimethylamine	(CH3)3N	SF6	1.055	0.279	N2O	0.3717	2.6390	V	6 to 24	Yes
Trimethylborate(TMB)	B(OCH3)3	SF6	0.769	0.203	SF6	0.2900	4.6388	V	1 to 5	Yes
Trimethylphosphite(TMP)	P(OCH3)3	SF6	0.599	0.158	SF6	0.3116	5.5393	V	1 to 5	Yes
Tritium	T2	N2	0.986	0.986	He	1.1709	0.2693		10 to 40	No
Tungsten Hexafluoride	WF6	SF6	0.815	0.215	SF6	0.0956	13.2900	KZ	3 to 12	Yes
Uranium Hexafluoride	UF6	SF6	0.748	0.198	SF6	0.0881	15.7000	KZ	3 to 12	Yes
Vinyl Bromide	CH2CHBr	C2H6	0.960	0.462	SF6	0.1242	4.7720	V	10 to 40	No
Vinyl Chloride	CH2CHCl	C2H6	0.995	0.478	N2O	0.2052	2.7880	V	10 to 40	No
Water Vapor	H2O	N2	0.815	0.815	N2	0.4458	0.8040	V	3 to 12	Yes
Xenon	Xe	Ar	1.000	1.415	SF6	0.0379	5.8580	V	10 to 40	No

NOTE : Conversion of controller to or from these gases may alter dynamic response or stability.

In accordance with SEMI Standard E12-91, Standard Pressure is defined as 760 mm Hg(14.7 psia) , Standard Temperature is defined as 0 degrees C.

Recommended Procedures to Avoid Contamination of Highly Reactive Gas System

Gas systems for saline and other highly reactive gases are extremely vulnerable to contamination. When oxygen, water vapor or other gases combine with highly reactive molecules, solid contaminants, such as silicon dioxide, form in the system and can cause numerous problems with measurement and control equipment. These particles can be transported in sub-micron sizes that pass through filters, or , can be formed by reactions in the plumbing downstream of the filter. Although proper use of filters reduces contamination, this potential problem exists throughout the entire system.

The three primary causes of contamination are leaks, impure purge gases, and improper purging procedures. These problems and their solutions are discussed below.

Leaks

Plumbing leaks cannot be tolerated in a pure gas system. It is commonly believed that if a gas line pressure is above that of the ambient, the surrounding atmosphere will be prevented from entering the system. However, for any flow through an orifice or leak path, back diffusion from the surrounding gas (air) takes place into the gas stream. The system can then become contaminated with oxygen or moisture. THEREFORE, AVOID ALL LEAKS. Check the system when new, after replacing any equipment in the plumbing circuit, and on a periodic schedule. A leak detector capable of detecting leaks in the range of less than 1×10^{-9} atm cc/sec of helium, is recommended.

Impure Purging Gases

Before and after a highly reactive gas system has been exposed to air, it is necessary to purge the system with an inert gas, i.e. N_2 or Ar with less than 10 ppm O_2 and water vapor.

Each time a transition is made from purge gas to process gas or process gas to purge gas, there is mixing of the two gases throughout the plumbing ; any oxygen or moisture in the purge gas reacts and can be left behind as contamination. The practice of purging process lines between runs gives two potential mixing events per run, and can cause a cumulative buildup or contamination. With leak-tight Plumbing it is not necessary to remove process gas from the system. Buildup resulting from repetitive purging can be avoided by not purging between runs.

PURGING IS NOT NECESSARY, NOR IS IT RECOMMENDED, AT ANY OTHER TIME

In some installations process gas is fed into a reaction chamber that can be exposed to air, thereby causing oxidation of residual gas in the gas supply line. As a preventative measure, installation of a shutoff valve in the process line, close to the reaction chamber, is recommended.

Proper Purge Procedure

The most effective method for removing process gases from process tubing is cycle purging. Cycle purging alternates evacuating the gas system and pressurizing the gas system with an inert purge gas. This procedure causes absorbed molecules and trapped gases to be removed much more quickly than straight purging. Contaminants trapped in plumbing dead space and micro cavities can communicate with the process stream to cause virtual leak. Cycle purging helps to purge these cavities. Al-though the application of vacuum to the gas system is very effective, a similar procedure can be adapted for atmospheric systems.

Vacuum Systems

Figure 1 is a picture of a typical reactive gas flow controller installation. To initiate purge procedure. Close Valves 2 and 2, and open Valve 1. Evacuate the line to base pressure. Close Valve 1 and open Valve 2 for one or two minutes to pressurize the controller with purge gas. Close Valve 3 and open Valve 1 to evacuate the controller back down to base pressure. Repeat the alternating application of evacuation and pressurization at least 20 times. When removing the flow controller, pressurize it with the purge gas and then close all 3 valves.

Atmospheric Systems

Although a vacuum pump is highly desirable, a similar procedure can be performed without a pump. An aspirator powered by a high rate of purge gas flow to exhaust can be used to create a sub atmospheric pressure in the plumbing system. If an aspirator is not available, cycling between atmospheric pressure and purge gas line pressure is better than simply flowing purge gas. Pressurize and depressurize the controller for about 5 minutes. And repeat the procedure at least 40 times, rather than the 20 used for vacuum systems.

CAUTIION
When highly reactive gases are being ebacuated by pumping, the pump should contain a nitrogen blanket above the oil level to dilute the gases and prevent air from being absorbed into the pump. Take extreme precautions to dispose of any unreacted process gas.
CONSULTYOURPLANTSAFETYENGINEER

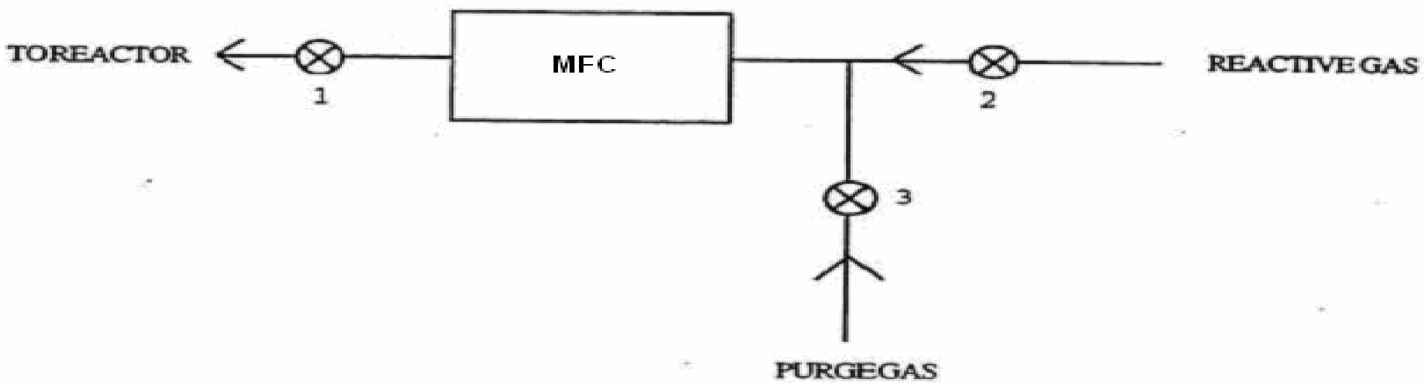


Figure 1

Highly Reactive Gas System

CERTIFICATE OF NON-HAZARDOUS CONTAMINATION

Return Acceptance Policy

In accordance with California Hazard Communication Regulation (General Industry Safety Orders Sec. 5194 Title 8, California Code CCR Title 22, California Hazardous Substances Information and Training Law and Department of Transportation requirements) Tylan General requires this Certificate of Non-Hazardous Contamination to be completed for all products received for service or repair. Complete and affix to the outside of the shipping container.

This form must be completed by a person with knowledge of all types of processes in which the equipment has been used. Non-completion of this form will cause delays and possible rejection of your equipment.

SEC. 1

Model ______________ Serial Number ______________

Has the equipment been used? Y N If Yes, go to section 2
If No, go to section 6

Sec. 2

List all substances, gases or by-products which have come in contact with the equipment.

Chemical/Substance:

____

____

____

Has the equipment been purged on-site? Y N If Yes, go to section 3
If no, go to section 4.

Sec. 3

Substance used in purge/flush process:

____

Sec. 4

Please attach all required Material Safety Data Sheets (MSDS) for all substances listed in Sec.2. Exposed equipment cannot be transported without the above safety information. Please contact your carrier for more specific instructions.

Sec. 5

The undersigned certifies that the composition of residue on the equipment listed above is of a non-hazardous nature and/or that the equipment has been purged or flushed on-site. All exposed equipment will be cleaned. Tylan General reserves the right to reject any equipment based on contamination resulting from exposure to severely toxic substances.

Sec. 6

Signed: ______________

Print Name: ______________

Title: ______________

Company: ______________

Telephone: ______________

Date of Signature: ______________